

ClassLens

Turning Mock-Exam Data into Personalised Student Study Plans a Cram School (or Tutor) Can Sell

Big Data Systems — Final Project (Spring 2026, National Taiwan University)

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GitHub repository: <https://github.com/petercechung/classlens>

Live demo: <https://classlens-demo.vercel.app> (replace with your Vercel URL, or remove)

Executive summary

Taiwanese cram schools (buxiban) run weekly mock exams and then discard the answer-level data after computing a raw score. **ClassLens** ingests that existing answer-sheet data and turns it into a **personalised study plan for each student**: the weak topics that are also heavily tested, ranked by payoff, with a two-week practice sequence and a **projected score lift** if the plan is followed — delivered as a print-ready report. Internal-efficiency framing (saving teacher time) is too weak a motivator on its own; the sellable value is the personalised recommendation, which a school uses to **recruit and retain** (showing parents exactly what their child should work on and the expected gain). The same per-student report works with no school at all, so an **independent tutor** can use it directly — which also seeds the first real data and demand evidence. The system does the routine diagnosis, recommendation and report-writing; the teacher keeps the teaching.

1. Target customer

Primary customer: owners / head teachers of **independent and small-chain, exam-focused cram schools serving junior- and senior-high students** — the segment that runs frequent mock exams and lives or dies on results and parent trust. **Beachhead / design partner: independent tutors** (including the author's own tutoring students), who need no procurement decision and let the product prove value on day one.

Why personalisation, not admin efficiency, is the wedge. Saving a teacher an hour is a weak purchase trigger — teacher time is cheap and the pain is mild. What a school will pay for is a **per-student personalised plan with a projected score gain**, because that is something it can put in front of parents to recruit and to justify staying enrolled. The deliverable is therefore a sales/retention asset, not an internal report.

Why this wedge. Taiwan has roughly **12,500 academic (wenli) cram schools (~17,500 operating entities)**, an industry estimated at **NT\$150–170 billion/year**, highly fragmented into tiny operators with budget but no data/IT team — the textbook SMB SaaS buyer. The junior/senior-high exam-prep slice is where exam stakes, parent willingness-to-pay and the value of diagnostics are all highest (the elementary majority is larger in headcount but lower in analytics need).

The job today. Teachers cull wrong answers and write parent feedback by hand, or rely on a raw score. Walled content platforms (PaGamO, Yincai Net) only analyse *their own* question banks, not the school's own mock exams. ClassLens analyses the data the school already owns — the unserved gap.

Secondary / expansion: small-chain head offices (one rollout covers many branches) and a later admissions-portfolio (xuexi licheng) analytics module. Students/parents are the data source and the value delivered, *not* the direct payer.

2. Evidence of demand and willingness to pay

2.1 Data-acquisition process

Following the Homework-2 methodology, demand is established from three sources:

- **Public / market data:** Ministry-of-Interior and industry statistics on cram-school counts, sector value and per-capita tutoring spend (below).
- **Competitor pricing benchmarks:** existing tools' published prices, used as willingness-to-pay anchors.
- **Practitioner interviews + survey (reproducible):** a 6-question structured survey and a 10-school interview guide (in the repo, survey/) targeting school owners via the author's tutoring / cram-school network — measuring minutes per exam spent on diagnosis/reporting and what they would pay to automate it.

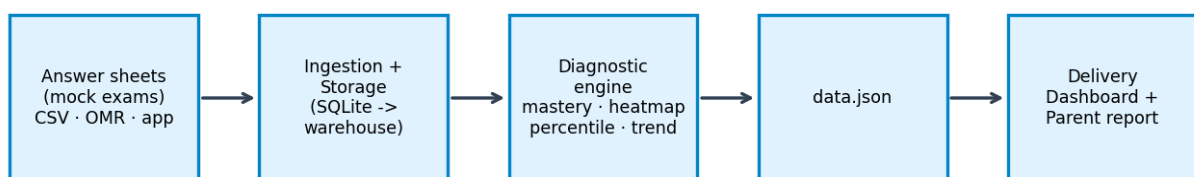
2.2 Quantified demand signals

Signal	Value	Source
Academic cram schools (Mar 2025)	~12,539 (≈17,500 entities)	PTS / Global Views
Sector value (estimate)	NT\$150–170 bn / yr	UDN / Business Weekly
Per-capita tutoring spend (JH, 2018)	≈ NT\$52,853 / student / yr	MOE Statistics
Comparable B2C tool WTP (HousePlus AVM)	freemium + paid	houseplus.com.tw
Analogue SaaS WTP band	US\$25–199 / mo	PropStream / Mashvisor

Willingness to pay (pricing model). B2B subscription priced per branch: **NT\$3,000–8,000 / month / branch**, or per enrolled student per term. The value is concrete: at NT\$45,000+ tuition per student-year, retaining a single student who would otherwise leave covers many months of subscription, and the parent report doubles as a recruitment tool. Comparable analytics/SaaS tools in adjacent verticals sustain US\$25–199/month, supporting this band.

3. System design

ClassLens is a four-stage pipeline: **ingestion** → **storage** → **processing** → **delivery**.



3.1 Data sources & ingestion (low friction)

Adoption hinges on ingesting the data a school *already has*. Schools do not keep clean long-format tables; they have a card-reader (OMR) or Excel export: one row per student, one column per question, cells = the chosen answer (or a correct/incorrect mark), plus a one-row answer key. The importer (`import_data.py`) converts exactly that into the pipeline's inputs. Crucially, **item difficulty is computed automatically** from the empirical pass rate, so the only one-time input is a question→topic map (or an annotated answer key). This removes the main barrier to a teacher trying it on an existing exam.

No content lock-in: ClassLens analyses the school's own exams (unlike walled platforms that require their question bank).

3.2 Storage

SQLite for the demo; the schema (students, items, responses) and the GROUP-BY aggregations are warehouse-agnostic. At scale, responses is the only large, append-only table, partitioned by (school, term).

3.3 Processing — the diagnostic engine

Mastery is **difficulty-weighted** (an IRT-style scheme): a correct answer on a hard item counts more, so the engine recovers a student's true weak topics rather than echoing the raw score. It computes topic mastery, a class weakness heatmap, cohort percentile and a term trend. The core output is a **personalised study plan**: topics are ranked by $gap \times exam-weight$ (a weak topic that is also heavily tested has the highest payoff), sequenced into a two-week plan, with a **projected score contribution** from raising each to an achievable target — so the report promises a concrete expected gain (e.g. 47.9 → 60.3). All metrics are explainable SQL aggregations + light Python: cheap, and trustworthy for teachers and parents.

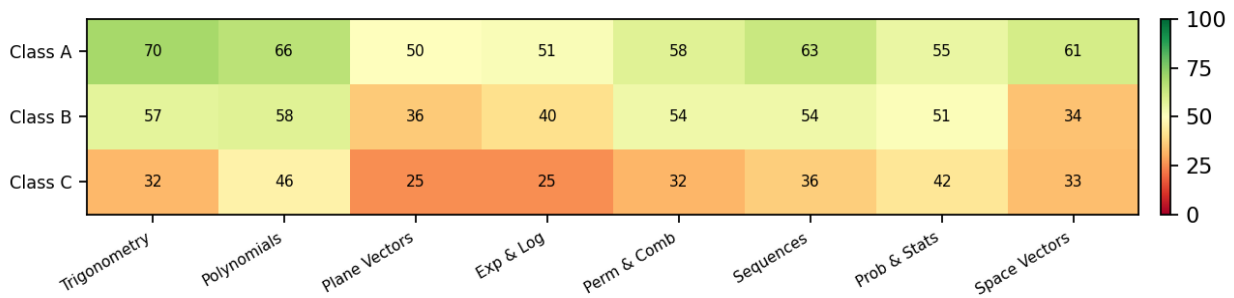


Figure: real pipeline output — class weakness heatmap across the three demo classes (mean mastery %, latest mock exam).

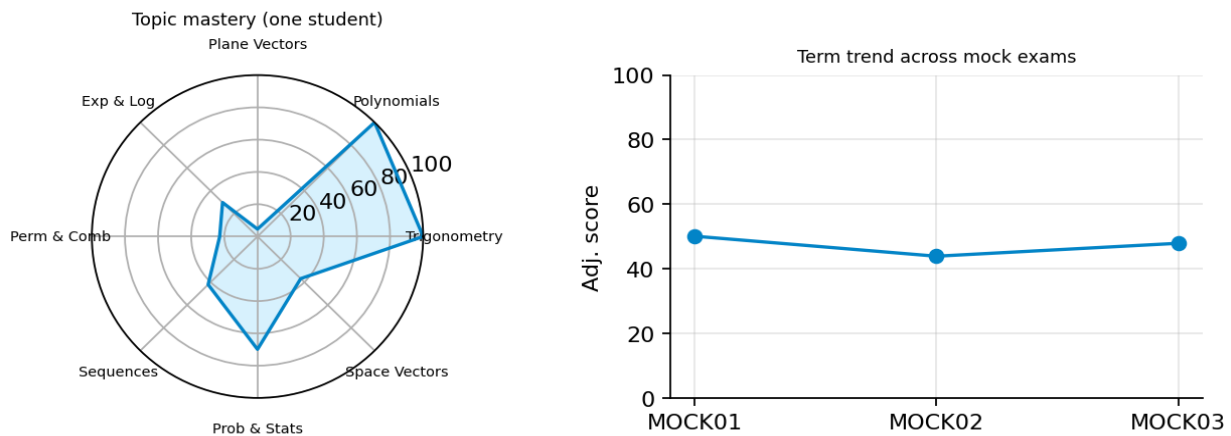


Figure: real pipeline output — one student's topic-mastery radar and term trend, as shown in the parent report.

3.4 Delivery

Two static artifacts that read data .json at runtime: a **teacher dashboard** (class heatmap, ranked students, weak topics) and a **printable parent report** (report.html?student=ID; radar, trend, strengths/weaknesses, targeted practice). Both deploy as a read-only static bundle (Vercel/Render/Pages).

3.5 Scalability & cost

The workload is bursty and batch-shaped: each exam upload triggers one diagnostic run, embarrassingly parallel by school/exam — no always-on streaming needed. At 100x (~100 branches, ~25k students, ~30M responses/term) the relational store moves to Postgres + object storage for raw scans; reports are pre-rendered on upload.

4. Go-to-market difficulties (bonus)

- **Trust & adoption:** diagnostics must be explainable and run on the school's own exams; we avoid black-box scores. Friction is minimised by importing an existing card-reader/Excel export (no data re-entry, difficulty auto-computed), so a free one-class, one-exam pilot takes minutes.

- **Data acquisition & privacy:** student data is sensitive under Taiwan's PDPA. Mitigations: school-as-data-controller, anonymised IDs, single-tenant/on-prem option, no data resale.
- **Cold start:** solved via the author's cram-school connections for a first design-partner school, seeding both demo data and demand evidence.
- **Competition & moat:** PaGamO / Yincai Net are content platforms (walled question banks, K-12 literacy, school channel). Our moat is data-infrastructure on the school's own materials + the parent-report workflow, not content. Large chains may build in-house, so we target independents and small chains first.
- **Unit economics:** near-zero marginal cost per added school (batch compute on small data); profitability is driven by branch count and retention, not heavy infrastructure.

References

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